

1) Write a Python program to read an entire text file

```
In [1]: 1 path = r'C:\velocity coding\assignments & solutions\Assignment solutions\Assignment_7_files&folders\deep_learning.txt'
2 with open(path) as f :
3     a = f.read()
4     print(a)
```

What is deep learning? Deep learning is a subset of machine learning, which is essentially a neural network with three or more layers. These neural networks attempt to simulate the behavior of the human brain—albeit far from matching its ability—allowing it to “learn” from large amounts of data.

```
In [ ]: 1
```

2) Write a Python program to read the first n lines of a file

```
In [6]: 1 path = r'C:\velocity coding\class files\05_17_File Handling_&_OS_Module\review.txt'
2 n = 7
3 with open(path) as f :
4     for i in range(n) :
5         a = f.readline()
6         print(a)
```

SO ASK YOURSELF WHAT " 8MM " (" EIGHT MILLIMETER ") IS REALLY ALL ABOUT .

IS IT ABOUT A WHOLESOME SURVEILLANCE MAN WHO LOSES SIGHT OF HIS VALUES AFTER BECOMING ENMESHED IN THE SEEDY , SLEAZY UNDERWORLD OF HARDCORE PORNOGRAPHY ?

IS IT ABOUT THE BUSINESS ITSELF , HOW , BUBBLING JUST BENEATH THE SURFACE OF BIG-TOWN AMERICANA , THERE'S A SORDID WORLD OF SICK AND DEPRAVED PEOPLE WHO WON'T NECESSARILY STOP SHORT OF MURDER IN ORDER TO SATISFY THEIR SICK AND TWISTED DESIRES ?

OR IS IT ABOUT THOSE WHO CAN , THOSE WHO ARE IN A POSITION TO INFLUENCE THE MAKING OF THE KINDS OF FILMS SICK AND DEMENTED PEOPLE WANT TO SEE ?

I'M NOT TALKING ABOUT SNUFF FILMS , SUPPOSED " DOCUMENTARIES " OF VICTIMS BEING BRUTALIZED AND KILLED ON CAMERA .

I'M TALKING ABOUT FILMS LIKE " 8MM " AND ITS DIRECTOR , JOEL SCHUMACHER .

WITH A RECENT RUN OF BIG BUDGET MOVIES TO HIS CREDIT-- " BATMAN & ROBIN , " " A TIME TO KILL , " " BATMAN FOREVER , " " THE CLIENT " --SCHUMACHER CERTAINLY HAS THAT KIND OF INFLUENCE .

```
In [ ]: 1
```

3) Write a Python program to append text to a file and display the text

```
In [12]: 1 path = r'C:\velocity coding\class files\05_17_File Handling_&_OS_Module\data_science.txt'
2 with open(path,'a') as f :
3     f.write(' It has a very good scope in the future')
4
5 with open(path) as t:
6     s = t.read()
7     print(s)
```

Python and Data Science can yield very lucrative salary It has a very good scope in the future

```
In [ ]: 1
```

4) Write a Python program to read the last n lines of a file

```
In [13]: 1 path = r'C:\velocity coding\class files\05_17_File Handling &_amp; OS_Module\review.txt'
2 n = 6
3 with open(path) as f:
4     l1 = f.readlines()
5     for i in range(len(l1)-n,len(l1)):
6         print(l1[i])
```

" 8MM " DOESN'T CONDEMN OR CONDONE ITS SUBJECT MATTER , IT JUST EXPLOITS IT .

THE IRONY , OF COURSE , IS THAT SCHUMACHER AND " SEVEN " SCRIBE ANDREW KEVIN WALKER'S VISION OF LIFE IN THE SNUFF LANE IS LIMITED BY WHAT THEY CAN SHOW IN AN R-RATED , FIRST-RUN HOLLYWOOD PRODUCT .

SO WE ONLY SEE SNIPPETS OF SNUFF , AND A LOT MORE FOOTAGE OF NICOLAS CAGE COVERING HIS FACE IN HORROR .

LATER IT'S THE TURN OF JOAQUIN PHOENIX (WHO'S QUITE GOOD AND BY FAR THE FILM'S MOST INTERESTING CHARACTER AS ADULT BOOKSTORE FLUNKY MAX CALIFORNIA) TO COVER HIS FACE AS THE HORRID THING IS SCREENED OVER AND OVER AGAIN .

ALL THIS TO GET TO THE FAMILIAR YET OFFENSIVE " REVELATION " THAT SEXUAL DEVIANTS ARE NOT , INDEED , MONSTERS BUT EVERYDAY PEOPLE LIKE YOU AND ME .

NEITHER SUPER NOR STANDARD , " 8MM " IS SHOCKING ONLY IN ITS BANALITY .

```
In [ ]: 1
```

5) Write a Python program to read a file line by line and store it into a list

```
In [5]: 1 path = r'C:\velocity coding\class files\05_17_File Handling &_amp; OS_Module\deep_learning3.txt'
2 l1 = []
3 with open(path) as f :
4     a = f.readline()
5     while a :
6         print(a)
7         l1.append(a)
8         a = f.readline()
9     print()
10    print(l1)
```

What is deep learning? Deep learning is a subset of machine learning,

which is essentially a neural network with three or more layers.

These neural networks attempt to simulate the behavior of the human

brain—albeit far from matching its ability—allowing it to “learn” from large amounts of data.

It is a subset of Machine Learning

['What is deep learning? Deep learning is a subset of machine learning, \n', 'which is essentially a neural network with three or more layers. \n', 'These neural networks attempt to simulate the behavior of the human \n', 'brain—albeit far from matching its ability—allowing it to “learn” from large amounts of data.\n', 'It is a subset of Machine Learning']

```
In [ ]: 1
```

6) Write a Python program to count the number of lines in a text file

```
In [16]: 1 path = r'C:\velocity coding\class files\05_17_File Handling &_amp; OS_Module\review.txt'
2 with open(path) as f :
3     a = f.readlines()
4     print(len(a))
```

27

```
In [ ]: 1
```

7) Write a Python program to count the frequency of words in a file

```
In [1]: 1 path = r'C:\velocity coding\assignments & solutions\Assignment solutions\Assignment_7_files&folders\deep_learning3.txt'
2 with open(path) as f :
3     a = f.read()
4     l1 = a.split()
5     s1 = set(l1)
6     for i in s1:
7         print(f'count of "{i}" in given text is {l1.count(i)}')
```

```
count of "layers." in given text is 1
count of "learning," in given text is 1
count of "amounts" in given text is 1
count of "data." in given text is 1
count of "network" in given text is 1
count of "a" in given text is 3
count of "deep" in given text is 1
count of "or" in given text is 1
count of "to" in given text is 2
count of "its" in given text is 1
count of "Machine" in given text is 1
count of "Deep" in given text is 1
count of "is" in given text is 4
count of "What" in given text is 1
count of "of" in given text is 4
count of "These" in given text is 1
count of "far" in given text is 1
count of "essentially" in given text is 1
count of "neural" in given text is 2
count of "networks" in given text is 1
count of "It" in given text is 1
count of "subset" in given text is 2
count of "attempt" in given text is 1
count of "brain-albeit" in given text is 1
count of "matching" in given text is 1
count of "human" in given text is 1
count of "machine" in given text is 1
count of "large" in given text is 1
count of "more" in given text is 1
count of "which" in given text is 1
count of "Learning" in given text is 1
count of "three" in given text is 1
count of "ability-allowing" in given text is 1
count of "with" in given text is 1
count of "learning?" in given text is 1
count of "the" in given text is 2
count of "it" in given text is 1
count of "learning" in given text is 1
count of "from" in given text is 2
count of "behavior" in given text is 1
count of "'learn'" in given text is 1
count of "simulate" in given text is 1
```

```
In [ ]: 1
```

8) Write a Python program to copy the contents of a file to another file

```
In [9]: 1 path = r'Assignment_7_files&folders\deep_learning3.txt'
2 new_path1 = r'Assignment_7_files&folders\deep_learning4.txt'
3 new_path2 = r'Assignment_7_files&folders\deep_learning5.txt'
4
5 with open(path) as f :
6     t = f.read()
7     with open(new_path1,'a') as f:
8         f.write(t)
9     with open(new_path2,'a') as f:
10        f.write(t)
```

```
In [ ]: 1
```

9) Write a Python script to display the various Date Time formats - Go to the editor

a) Current date and time . b) Current year . c) Month of year

d) Week number of the year . e) Weekday of the week . f) Day of year

g) Day of the month . h) Day of week

```
In [15]: 1 import datetime
2 current_date = datetime.datetime.now().date()
3 print(current_date)
4 current_time = datetime.datetime.now().time()
5 print(current_time)
6 current_year = datetime.datetime.now().year
7 print(current_year)
8 current_month = datetime.datetime.now().month ##month of the year
9 print(current_month)
```

2023-06-01
19:57:49.767770
2023
6

```
In [26]: 1 a = datetime.datetime.today()
2 b = datetime.datetime.strftime(a, '%B') #month name
3 print(b)
```

June

```
In [27]: 1 a = datetime.datetime.today()
2 b = datetime.datetime.strftime(a, '%U') #week number of the year
3 print(b)
```

22

```
In [30]: 1 a = datetime.datetime.today()
2 b = datetime.datetime.strftime(a, '%A') #weekday name
3 print(b)
```

Thursday

```
In [32]: 1 a = datetime.datetime.today()
2 b = datetime.datetime.strftime(a, '%j') #day number of the year
3 print(b)
```

152

```
In [33]: 1 a = datetime.datetime.today()
2 b = datetime.datetime.strftime(a, '%d') #day number of the month
3 print(b)
```

01

```
In [35]: 1 a = datetime.datetime.today()
2 b = datetime.datetime.strftime(a, '%w') #day number of the week
3 print(b)
```

4

```
In [ ]: 1
```

10) Write a Python program to determine whether a given year is a leap year

```
In [44]: 1 y = eval(input('Enter the year in YYYY format : '))
2 a = datetime.datetime(y,12,31)
3 x = datetime.datetime.strftime(a, '%j')
4 print('It is a leap year') if x == '366' else print('It is not a leap year')
```

Enter the year in YYYY format : 2024
It is a leap year

```
In [ ]: 1
```

11) Write a Python program to convert a string to datetime. Go to the editor

Sample input String : Jan 1 2014 2:43PM

```
In [48]: 1 str1 = 'Jan 1 2014 2:43PM'
2 a = datetime.datetime.strptime(str1, '%b %d %Y %I:%M%p')
3 print(a)
```

2014-01-01 14:43:00

```
In [ ]: 1
```

12) Write a Python program to subtract five days from current date

```
In [49]: 1 a = datetime.datetime.today()
2 b = a + datetime.timedelta(days = -5)
3 print(b)
```

2023-05-27 21:19:27.860175

```
In [ ]: 1
```

13) Write a Python program to print yesterday, today, tomorrow

```
In [51]: 1 a = datetime.datetime.today()
2 for i in range(-1,2):
3
4     b = a + datetime.timedelta(days = i)
5     print(b)
```

2023-05-31 21:35:54.371583

2023-06-01 21:35:54.371583

2023-06-02 21:35:54.371583

```
In [ ]: 1
```

14) Write a Python program to print next 5 days starting from today

```
In [53]: 1 a = datetime.datetime.today()
2 for i in range(6):
3
4     b = a + datetime.timedelta(days = i)
5     print(b)
```

2023-06-01 21:37:13.908032

2023-06-02 21:37:13.908032

2023-06-03 21:37:13.908032

2023-06-04 21:37:13.908032

2023-06-05 21:37:13.908032

2023-06-06 21:37:13.908032

```
In [ ]: 1
```

15) Write a Python program to drop microseconds from DateTime

```
In [62]: 1 a = datetime.datetime.today()
2 b = datetime.datetime.strftime(a, '%Y-%m-%d %X')
3 print(b,type(b))
4 c = datetime.datetime.strptime(b, '%Y-%m-%d %H:%M:%S')
5 print(c,type(c))
```

2023-06-01 21:45:32 <class 'str'>

2023-06-01 21:45:32 <class 'datetime.datetime'>

```
In [ ]: 1
```

16) Write a Python program to find the date of the first Monday of a given week

```
In [78]: 1 a = int(input('Enter week number of the year : '))
2 b = input('Enter the year in the format YYYY : ')
3 d = datetime.datetime.strptime(b, '%Y')
4 c = d + datetime.timedelta(weeks = a-1)
5 for i in range(7):
6     e = c + datetime.timedelta(days = i)
7     if datetime.datetime.strftime(e, '%A') == 'Monday':
8         print(e.date())
9         break
```

Enter week number of the year : 15

Enter the year in the format YYYY : 2023

2023-04-10

```
In [ ]: 1
```

17) Write a Python program to select all the Sundays of a specified year

```
In [89]: 1 b = input('Enter the year in the format YYYY : ')
2 d = datetime.datetime.strptime(b, '%Y')
3 for i in range(7):
4     e = d + datetime.timedelta(days = i)
5     if datetime.datetime.strftime(e, '%A') == 'Sunday':
6         break
7 for i in range(54):
8     f = e + datetime.timedelta(weeks = i)
9     if datetime.datetime.strftime(f, '%Y') == b:
10        print(f)
```

Enter the year in the format YYYY : 2012

2012-01-01 00:00:00
2012-01-08 00:00:00
2012-01-15 00:00:00
2012-01-22 00:00:00
2012-01-29 00:00:00
2012-02-05 00:00:00
2012-02-12 00:00:00
2012-02-19 00:00:00
2012-02-26 00:00:00
2012-03-04 00:00:00
2012-03-11 00:00:00
2012-03-18 00:00:00
2012-03-25 00:00:00
2012-04-01 00:00:00
2012-04-08 00:00:00
2012-04-15 00:00:00
2012-04-22 00:00:00
2012-04-29 00:00:00
2012-05-06 00:00:00
2012-05-13 00:00:00
2012-05-20 00:00:00
2012-05-27 00:00:00
2012-06-03 00:00:00
2012-06-10 00:00:00
2012-06-17 00:00:00
2012-06-24 00:00:00
2012-07-01 00:00:00
2012-07-08 00:00:00
2012-07-15 00:00:00
2012-07-22 00:00:00
2012-07-29 00:00:00
2012-08-05 00:00:00
2012-08-12 00:00:00
2012-08-19 00:00:00
2012-08-26 00:00:00
2012-09-02 00:00:00
2012-09-09 00:00:00
2012-09-16 00:00:00
2012-09-23 00:00:00
2012-09-30 00:00:00
2012-10-07 00:00:00
2012-10-14 00:00:00
2012-10-21 00:00:00
2012-10-28 00:00:00
2012-11-04 00:00:00
2012-11-11 00:00:00
2012-11-18 00:00:00
2012-11-25 00:00:00
2012-12-02 00:00:00
2012-12-09 00:00:00
2012-12-16 00:00:00
2012-12-23 00:00:00
2012-12-30 00:00:00

In []: 1

18) Write a Python program to create a file and write some text and rename the file name

```
In [91]: 1 import os
2 text = 'Aabra ka Daabra'
3 path = r'Assignment_7_files&folders\deep_learning6.txt'
4 new_path = r'Assignment_7_files&folders\deep_learning_new.txt'
5 with open(path, 'w') as f :
6     f.write(text)
7
8 os.rename(path, new_path)
```

```
In [ ]: 1
```

19) What are different methods available in the OS module for creating a directory?

```
In [ ]: 1 1) os.mkdir() - To create a single empty directory
2 2) os.makedirs() - To create a tree structure of directory(one directory inside another and so on)
```

```
In [ ]: 1
```

20) Explain os.listdir() method.

```
In [ ]: 1 This will give you a list of all the files and folders present inside the given path.
2 Path will be passed as an argument inside the brackets
```

```
In [ ]: 1
```

21) What are different methods for removing directories and files in Python?

```
In [ ]: 1 1) os.remove()
2 This will remove one file at a time and the file path must be passed as an argument
3 2) os.rmdir()
4 This will delete an empty directory. Path of that empty directory should be passed as an argument
5 3) shutil.rmtree()
6 This will delete even non empty directory. Used to delete tree structured directory.
7 Path must be passed as an argument
```

```
In [ ]: 1
```

22) What are exceptions in Python?

```
In [ ]: 1 Exceptions are nothing but the errors that we encounter during the execution of the code. We can handle
2 runtime errors by using try - except block and this way of handling errors is also called exception handling.
3 We can not handle Syntax error and Indentation error using exception handling(i.e. by using try - except block)
```

```
In [ ]: 1
```

23) What are Built-in exceptions?

```
In [ ]: 1 Python Exceptions that are thrown by the standard library routines or by the interpreter itself are called built-in
2 exceptions which implies that they are present in the source code at all times. When an error occurs, Python raises
3 built-in exceptions by itself. There are a number of built in exceptions, for ex : NameError, TypeError, ValueError, etc
```

```
In [ ]: 1
```

24) What are User-defined Exceptions?

```
In [ ]: 1 User defined exceptions can be implemented by raising an exception explicitly, by using assert statement or by
2 defining custom classes for user defined exceptions.
3
4 Creating user defined Exception Class :
5 Our Exception Class should Implement Exceptions to raise exceptions. In this exception class, we can either put
6 pass or give an implementation. Let us define init function :
7
8 Code:
9 class JustException(Exception):
10 def __init__(self, message):
11 print(message)
12
13 In the example above, JustException is a user-defined Exception. This Class implements the Exception class. We have
14 defined the __init__ function that takes the message as a parameter and prints the message. This JustException Class
15 can implement everything like a normal class in Python.
16
17 Raising User-defined Exception :
18 When a certain condition is met, we use the raise keyword to throw an exception. To raise a User-defined Exception,
19 we can do the following:
20
21 Code:
22 raise JustException("Raise an Exception")
23
```

```
In [ ]: 1
```

25) When would you not use try-except?

```
In [ ]: 1 We use try - except block for risky code, i.e. when there is a probability that our written code can give some error.
2 So basically we are using try - except block to handle errors(for exception handling). So, if we are sure about
3 our code and if we know that it will not give any error, in that case we do not need to use any try-except block.
```

```
In [ ]: 1
```

26) Can try-except catch the error if a file can't be opened?

```
In [ ]: 1 If a file is not able to open then it will give some kind of Runtime error like FileNotFoundError, PermissionError,
2 OSError, etc. We can handle Runtime errors using try - except block. So yes even in this case we can catch the error
3 using try - except block. There are only two scenarios where try - except block can not handle errors and that is in
4 case of Syntax Error and Indentation Error.
5 If you want to see what kind of error occurred then there are two ways :
6 1) Using traceback library - using print(traceback.print_exc()) statement
7 2) Writing code in except block like :
8 except Exception as err:
9 print(err)
```

```
In [ ]: 1
```

***** Assignment 7 Completed *****

```
In [ ]: 1
```